

But most of them are free anyway. So you won't have to spend your Gandhis on buying a license to use the language. Also, you will find extensive documentation online, which is also maintained by a large community. This helps you to pick up the language easily and avoid missteps which other users might have faced. You will also find several web services where you can start coding in an online sandbox, so writing incorrect code won't burn down your home or office.

One thing which any beginner should be aware is that, not all programming languages are made alike. Some are generic and will allow you to do everything, while some of them are extremely specialised. Languages like Python, Swift and Ruby are mostly languages which you use to develop your backend. This is the code you will be using to run your project. Some languages like HTML, CSS are purely frontend languages and are used to develop interfaces.

While you wouldn't particularly need to develop an interface for your hardware project, if it is a software based one, you will need a good interface for the users.

For instance, to say 'Hello Digit' in various languages, you'll have to use a different syntax for each language:

```
>print "Hello Digit"
>printf "Hello Digit"
>say "Hello Digit"
>cout<<"Hello Digit"
>System.out.print("Hello Digit");
```

While the syntax might be different, each language is similar to another in terms of the function. So it is best to learn one language to a good extent, before trying to pick up other languages.

Hardware simulators and circuit designers

When designing hardware for your project, you may not have the necessary parts at hand or you'd want to experiment with different types of hardware. This is not possible every time as you may not have access to all the different types of hardware. It is also very risky to try out new circuits using prototyping boards, which are often expensive. So, it is best to try out the circuits online before actually using the hardware.

There are several online simulators which you can use, ranging from simple circuit designers to analysers, which can tell you every technical aspect of your project. The simulators are also dependent on the hardware